

Name: _____ / 25

Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

There are 6 questions worth 25 points on this quiz. No aids (book, calculator, etc.) are permitted.
Show all work for full credit.

1. [6 points] Given functions $f(x) = x^3 - 3x + 1$ and $g(x) = 2^x$, compute the following and **simplify** your answers:

a. $f(2) + g(2)$

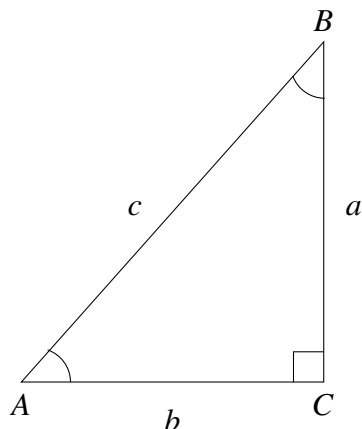
b. $f(0) - g(0)$

c. $g(f(1))$

2. [3 points] Write an equation of the line passing through the points $(1, 2)$ and $(-1, 5)$.

3. [3 points] Find all x -intercepts for the function $f(x) = 2x^2 - 7x + 5$.

4. [4 points] In the right triangle below, suppose $a = 5$ and $c = \sqrt{34}$.



Calculate the following:

$$\sin(A) =$$

$$\cos(A) =$$

$$\tan(A) =$$

5. [3 points] Evaluate the following:

a. $\sin\left(\frac{\pi}{4}\right) =$

b. $\tan\left(\frac{5\pi}{6}\right) =$

c. $\sec(\pi) =$

6. [6 points] State the domain and range of the following functions. Use interval notation.

a. $h(x) = \ln(x)$

domain: _____

range: _____

b. $g(x) = 3\cos(x) + 1$

domain: _____

range: _____

c. $f(x) = -\sqrt{x+1} - 5$

domain: _____

range: _____